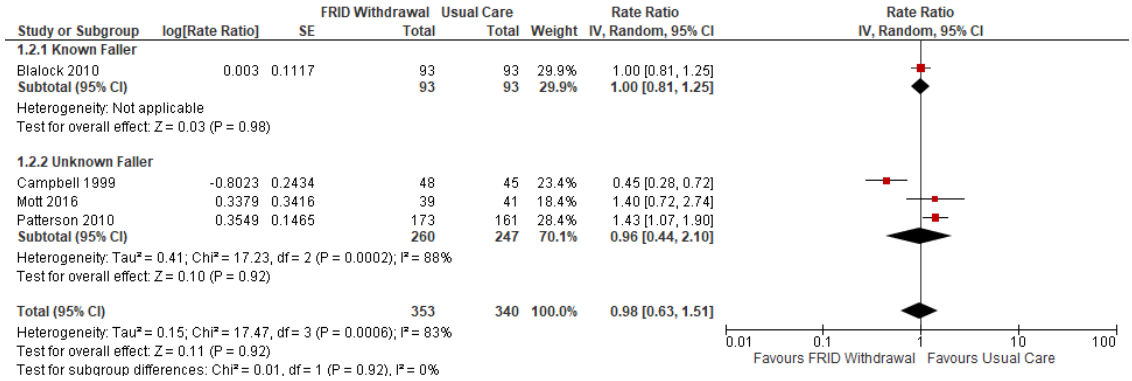
Supplementary Figure S1: OVID Medline Search Strategy

Database(s): OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:

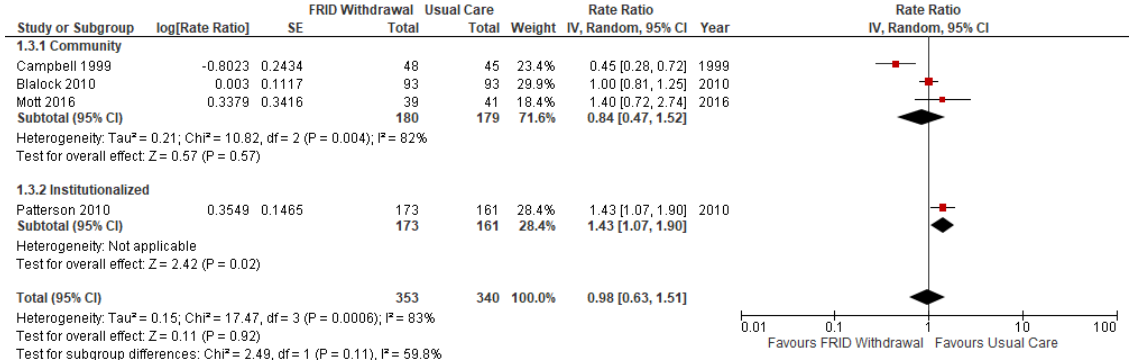
#	Searches
1	exp Accidental Falls/pc [Prevention & Control]
2	fall.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]
3	falls.mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]
4	exp Deprescriptions/
5	((medicat* or drug*) adj3 (deprescrib* or withdraw* or cessat* or stop* or discontin*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]
6	((antihypertensive* or diuretic* or beta-blocker* or sedative* or hypnotic* or neuroleptic* or antipsychotic* or antidepressant* or benzodiazepine* or narcotic* or opioid* or narcotic* or NSAID*) adj3 (deprescrib* or withdraw* or cessat* or stop* or discontin*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]
7	fall-risk increasing drugs.mp.
8	FRID.mp.
9	((medicat* or drug*) adj3 (review* or improv* or program*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]
10	exp "Drug-Related Side Effects and Adverse Reactions"/pc [Prevention & Control]
11	exp Medication Therapy Management/ or exp "Drug Utilization Review"/
12	4 or 5 or 6 or 7 or 8 or 9 or 10 or 11
13	1 or 2 or 3
14	12 and 13
15	remove duplicates from 14
16	exp Clinical Trial/
17	(randomized or randomised).ab,ti.
18	placebo.ab,ti.
19	randomly.ab,ti.
20	groups.ab,ti.
21	randomized controlled trial.pt.
22	controlled clinical trial.pt.
23	16 or 17 or 18 or 19 or 20 or 21 or 22
24	15 and 23

Supplementary Figure S2: Subgroup Analyses

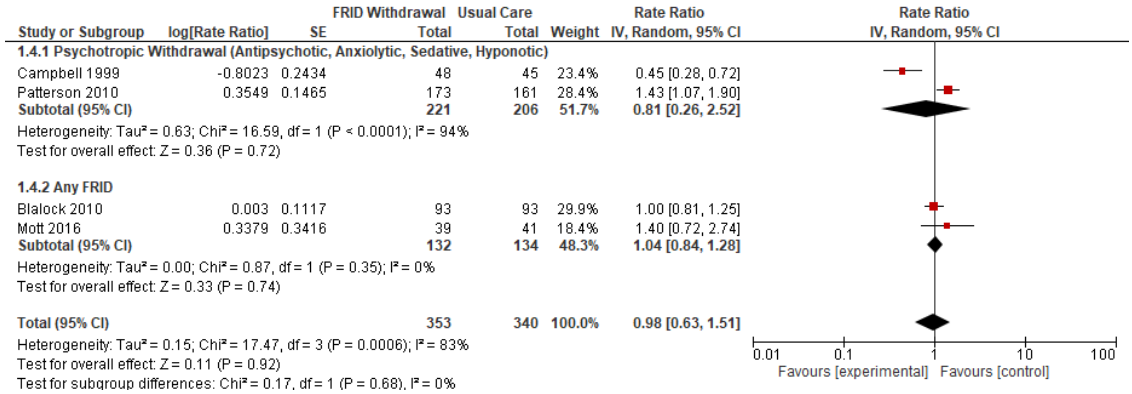
1.2 Falls Rate - Known vs. Unknown Faller



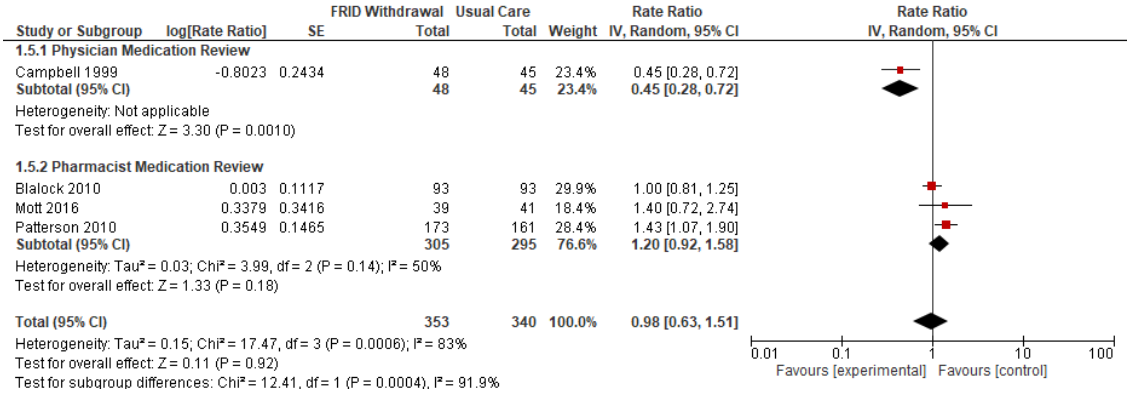
1.3 Falls Rate - Community vs. Institutionalized



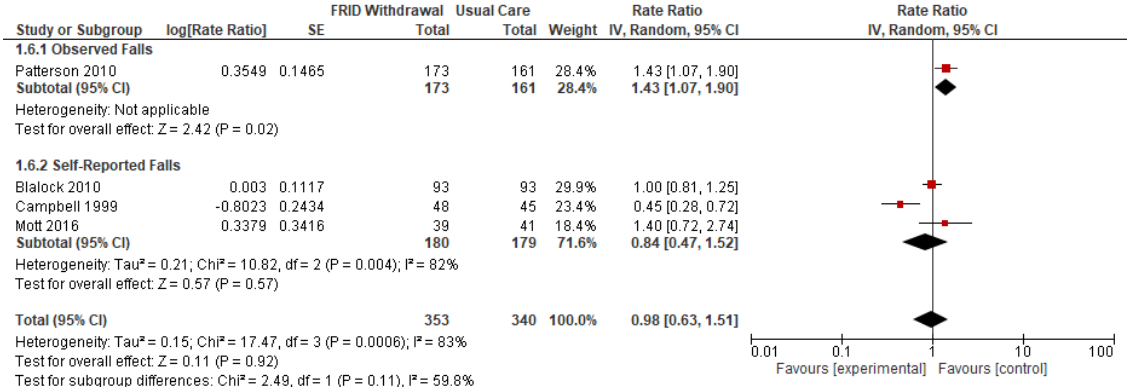
1.4 Falls Rate - Psychotropic Withdrawal vs. Any FRID Withdrawal



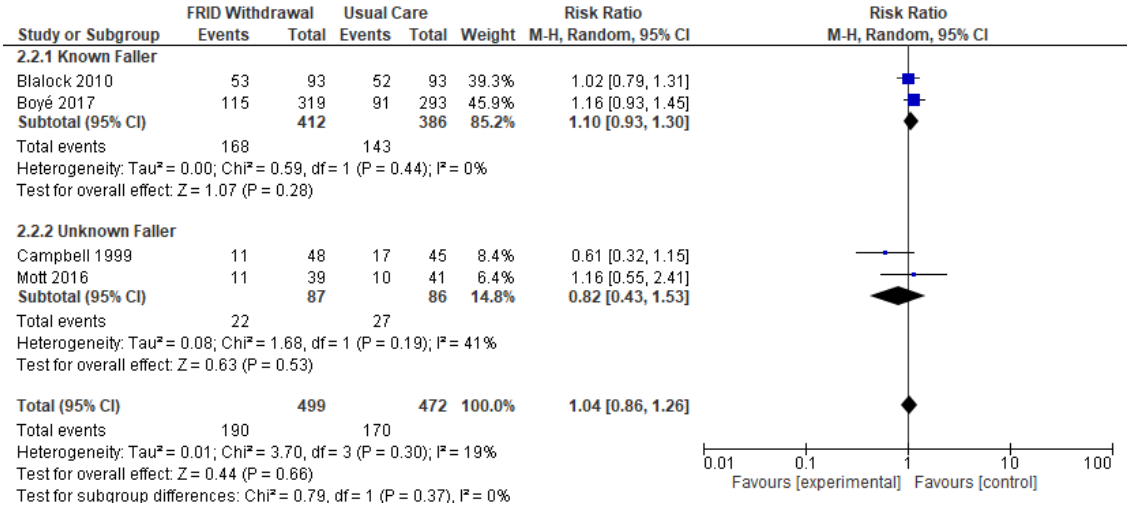
1.5 Falls Rate - Physician vs. Pharmacist Medication Review



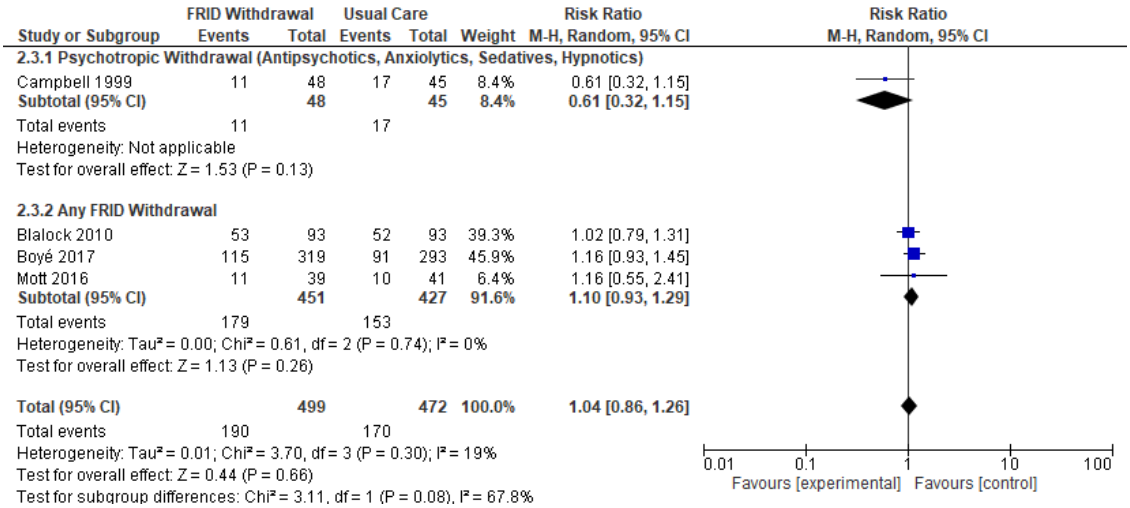
1.6 Falls Rate - Observed vs. Self-Reported Falls



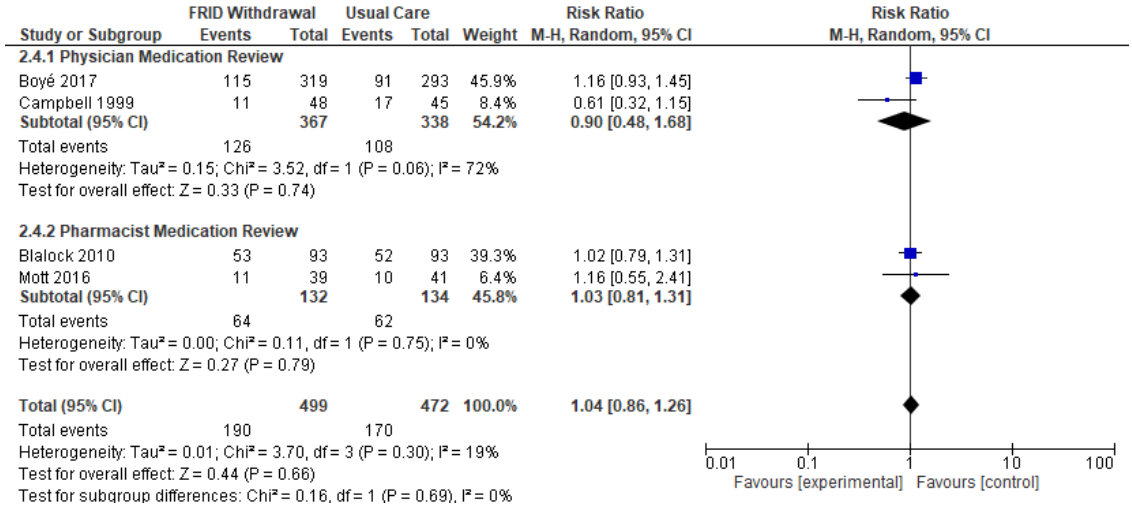
2.2 Falls Incidence - Known vs. Unknown Faller



2.3 Falls Incidence - Psychotropic Withdrawal vs. Any FRID Withdrawal



2.4 Falls Incidence - Physician vs. Pharmacist Medication Review



Supplementary Table S1: Subgroup Credibility Assessment – Clinician Medication Review

Physician vs. Pharmacist Medication Review Subgroup for Falls Rate

Design	**Criteria Met?**
Is the subgroup variable a characteristic measured at baseline or after randomization?	Yes – Variable determined at baseline
Is the effect suggested by comparisons within rather between studies?	No – Comparison between studies
Was the hypothesis specified a priori?	Yes
Was the direction of the subgroup effect specified a priori?	No
Was the subgroup effect one of a small number of hypothesized effects tested?	Yes – 1 of 5 analyses
Analysis	
Does the interaction test suggest a low likelihood that chance explains the apparent subgroup effect?	Yes – p =0.0004
Is the significant subgroup effect independent?	Yes
Context	
Is the size of the subgroup effect large?	Yes – RaR 0.45 vs. 1.20
Is the interaction consistent across studies?	No
Is the interaction consistent across closely related outcomes within the study?	No – Subgroup interaction was not seen for incidence of falls
Is there indirect evidence that supports the hypothesized interaction (biological rationale)?	No - No compelling external evidence supporting subgroup hypothesis

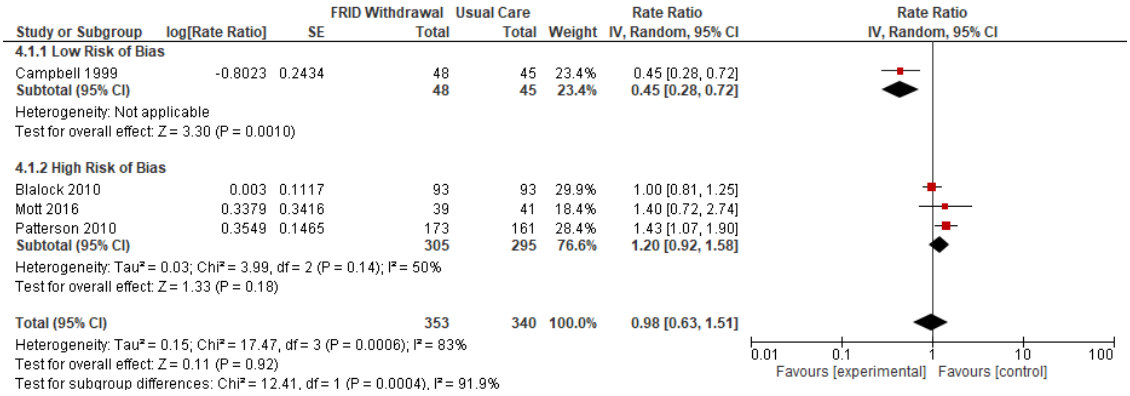
Supplementary Table S2: Subgroup Credibility Assessment – FRID Withdrawal Type

Antipsychotic vs. Any FRID Withdrawal for Falls Incidence

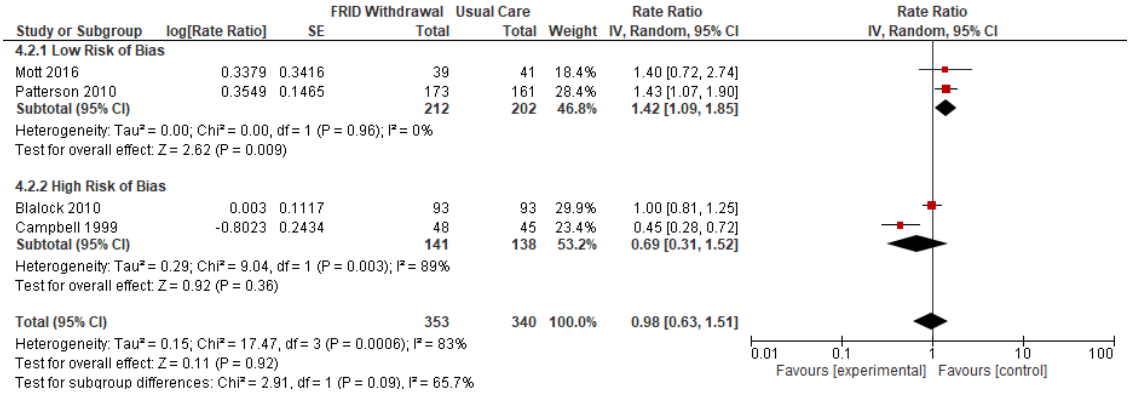
Design	**Criteria Met?**
Is the subgroup variable a characteristic measured at baseline or after randomization?	Yes – Variable determined at baseline
Is the effect suggested by comparisons within rather between studies?	No – Comparison between studies
Was the hypothesis specified a priori?	Yes
Was the direction of the subgroup effect specified a priori?	No
Was the subgroup effect one of a small number of hypothesized effects tested?	Yes – 1 of 3 analyses
Analysis	
Does the interaction test suggest a low likelihood that chance explains the apparent subgroup effect?	Yes – p=0.06
Is the significant subgroup effect independent?	No
Context	
Is the size of the subgroup effect large?	Yes – RR 0.61 vs. 1.14
Is the interaction consistent across studies?	No
Is the interaction consistent across closely related outcomes within the study?	No – Subgroup interaction was not seen for rate of falls
Is there indirect evidence that supports the hypothesized interaction (biological rationale)?	Yes – Antipsychotics associated with one of highest risks of falls. The withdrawal of any FRID may involve withdrawal of those with lower risks and limit potential benefit.

Supplementary Figure S3: Sensitivity Analyses

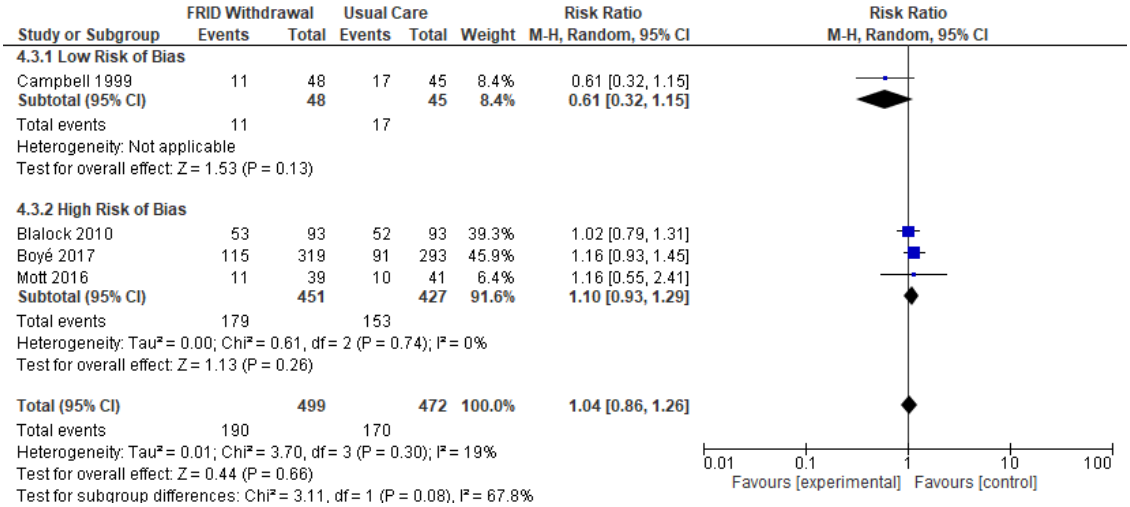
4.1 Falls Rate - Low vs. High Risk of Bias due to Blinding



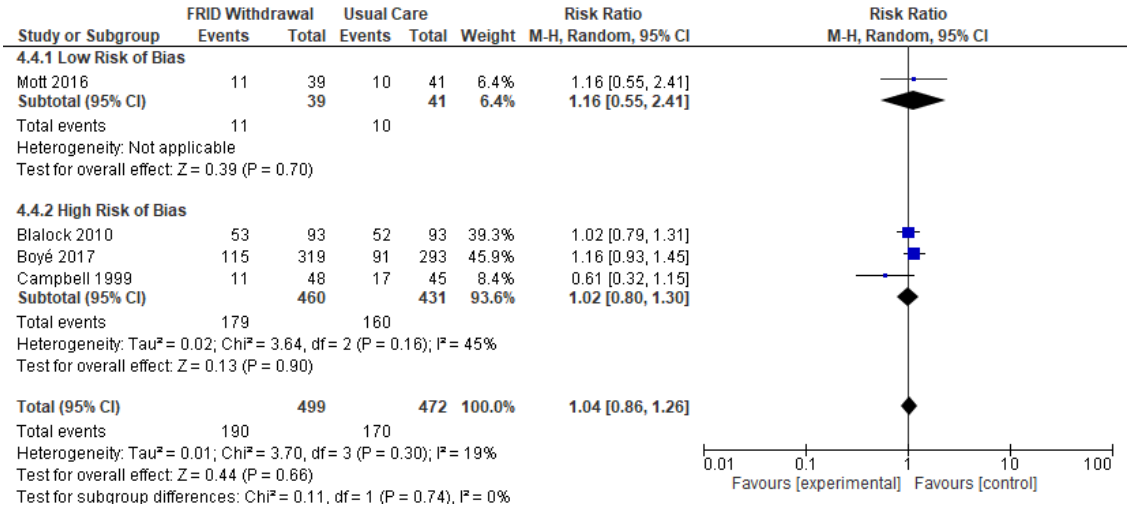
4.2 Falls Rate - Low vs. High Risk of Bias due to Attritional Bias



4.3 Falls Incidence - Low vs. High Risk of Bias due to Blinding

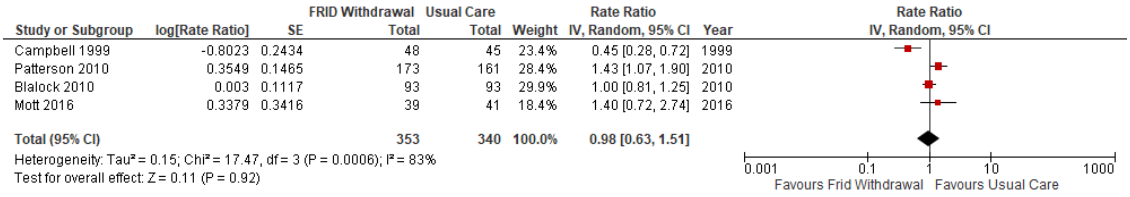


4.4 Falls Incidence - Low vs. High Risk of Bias due to Attrition Bias

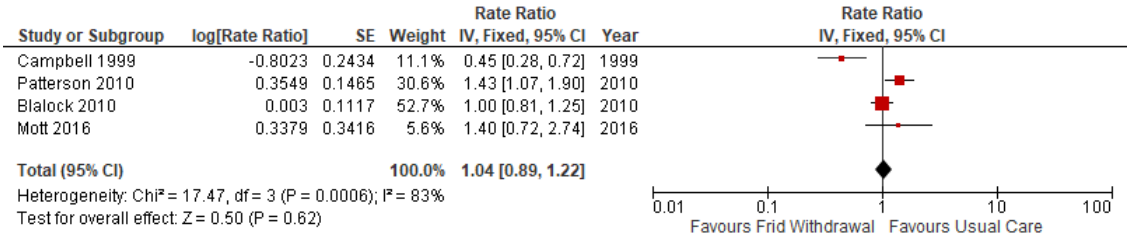


4.5 Falls Rate – Random vs. Effects Model

Random Effects Model

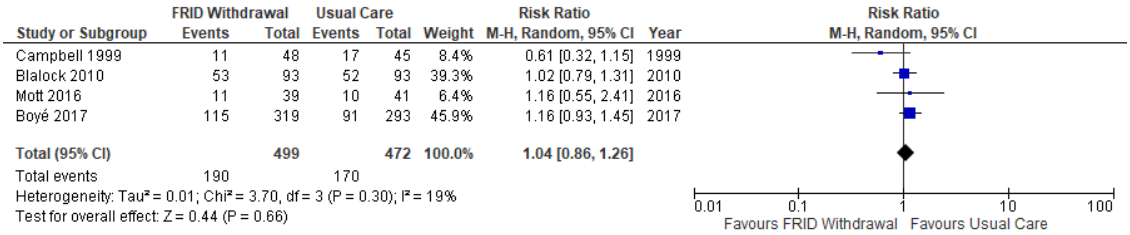


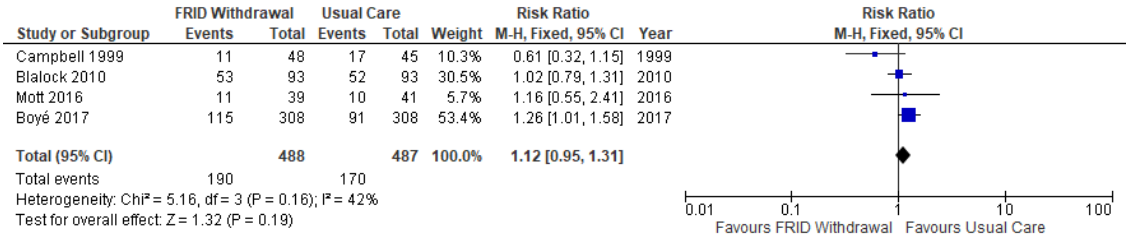
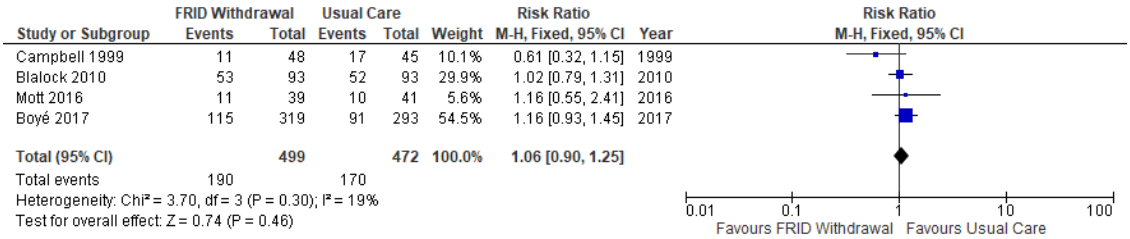
Fixed Effects Model



4.6 Falls Incidence – Random vs. Fixed Effects Model

Random Effects Model





Fixed Effects Model